

(a) mirroring, reference case (256^3 , 4 GPUs): 220 ms step

GPU 13 H2D 26



GPU busy 6% (13 ms)

largest source: 12 full-field syncs
420 MB/step, consumer reads ~5 MB

(b) managed vs explicit residency, production case (472M, 16 GPUs)

managed
memory



7.9x



explicit
residency

0.157 s, zero $O(N^3)$ PCIe crossings
(~260 KB packet + halo planes)

